



Filmon Misgane

Applied Mathematics | Data Science | Machine Learning

Mobile: (+47) 94725457

E-mail: filmontesfamikal@gmail.com

Citizenship: Norwegian



Profile

I am completing my master's degree in Mathematics for Applications this summer at the University of Oslo, following a bachelor's degree in Mathematics and Computer Science. My background combines strong quantitative training with practical experience in Python-based data analysis, modelling, machine learning, and development of computational tools for complex scientific problems. Through my thesis and research projects, I work with data preparation, structured analysis workflows, model development, validation, and clear communication of results. I am motivated by applying quantitative methods, machine learning, and analytical problem-solving to transform complex data into robust models, useful insight, and decision support.



Education

Present
↑
08.2024

MSc. Mathematics for Applications

University of Oslo (UiO)

- Master thesis: Numerical Methods for Nonlocal Conservation Laws.
- Designed and implemented Python solvers for first- and second-order numerical schemes, including analysis, testing, validation, and documentation.
- Developed strong competence in mathematical modelling, algorithm development, and structured quantitative problem-solving.

06.2024
↑
08.2021

BSc. Mathematics and Computer Science

University of Oslo (UiO)



Work Experience

Present
↑
04.2026

Summer Researcher

UiO Department of Geosciences in collaboration with SINTEF

- Working on scientific machine learning methods for subsurface energy storage, with a focus on Physics-Informed Neural Networks (PINNs) and Neural Operators.
- Contributing to data preparation, model development, validation, and quantitative analysis for PDE-based subsurface flow and transport problems.
- Using Python-based tools for data processing, machine learning, and evaluation of modelling approaches.
- Collaborating in an interdisciplinary research setting and communicating analytical results in a structured way.



GitHub

<https://github.com/Filmon492>



Skills

- Programming: Python, C++, MATLAB
- Data analysis and modelling: Pandas, NumPy, quantitative analysis, model development, model evaluation, data processing
- Machine learning: PINNs, Neural Operators, machine learning workflows for scientific applications
- Tools: Git/GitHub, PyTest, FEniCSx, FreeFEM, SymPy



Job References

Professor Ulrik Skre Fjordholm
Department of Mathematics, University of Oslo
E-mail: ulriksf@math.uio.no
Phone: (+47) 22 85 60 29
Relationship: Master's thesis supervisor



Languages

Norwegian	Fluent
-----------	--------

English	Fluent
---------	--------

Present ●
↑
04.2026

Summer Intern

DSB group, Department of Informatics

- Contributing to a software pipeline for medical imaging involving data processing, reconstruction, evaluation, and visualization.
- Supporting quantitative assessment of reconstruction methods and development of analysis workflows.
- Working with structured problem-solving and software development in an applied technical environment.

Present ●
↑
08.2022

Mentor

ENT3R, University of Oslo (UiO)

- Supporting high school students in mathematics and science, and communicating complex quantitative topics in a clear, structured, and accessible way.
-